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Socioeconomic Determinants of Cesarean Section Delivery in Bangladesh: A Survey-Weighted Statistical and Explainable Machine Learning Analysis.

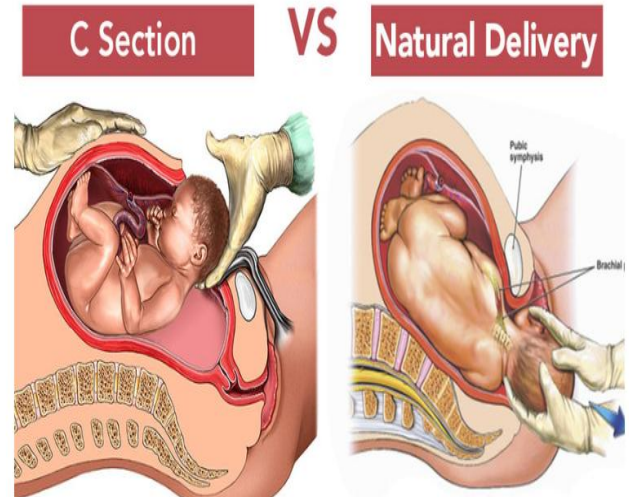
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What is C-Section & Why Does It Matter?

Life-saving surgery

Cesarean section is a surgical procedure to deliver a baby through incisions in the abdomen and uterus.

Global rates have doubled

Global C-section rates have risen from 21% and are expected to reach nearly 30% by 2030, exceeding the WHO recommended level of 10–15%.

Over-use causes harm

Non-indicated C-sections raise maternal morbidity, neonatal respiratory distress, long-term reproductive risks & healthcare costs.

**WHO
Recommends:**
10–15%

Global Rate

Now:

>21%

**& rising to 29–30%
by 2030**

**3-4 ×
Excess above WHO limit**

C-Section Rate in Bangladesh over time

Among the highest in any low-or-middle-income country

Key Facts

44.65% C-section rate (2022) — among highest in LMICs

84% Private facility C-section rate

~35.6% C-section rate in Public deliveries

21% + Global C-Section rate

Early 2000s

<5%

BDHS 2014

~31%

BDHS 2017

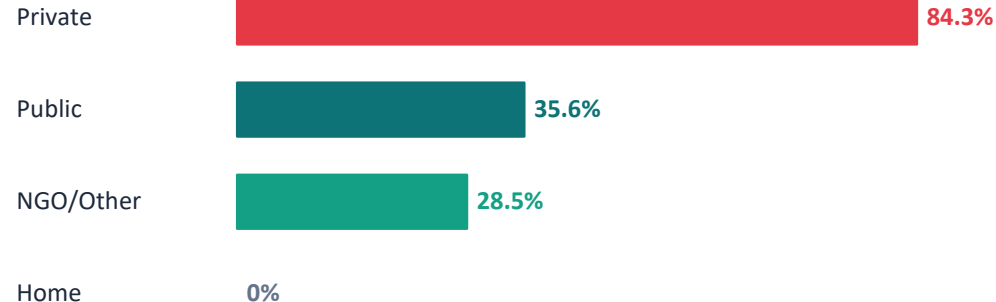
~35%

BDHS 2022

44.65%

Trend →

The Hidden Divide: Private vs. Public Facilities



C-section rate by facility type →

Research Gaps and Our Study Contributions in BDHS 2022

What Studies Are Missing

- 1 Existing studies mostly use traditional regression
- 2 Limited ML applications to survey-weighted BDHS data
- 3 Limited model interpretability
- 4 Lack of integrated analytical framework

Our Study Addresses All of This

- ✓ Apply survey-weighted GEE for population-representative results
- ✓ Multiple ML classifiers with survey-weighted BDHS data
- ✓ Apply SHAP-based Explainable AI for explanations
- ✓ Combines Statistics + ML + XAI

Research Objectives Of Our Study



Primary Objective

To investigate the socioeconomic and healthcare factors influencing cesarean section (C-section) delivery in Bangladesh using BDHS 2022 data.



Specific Objectives

- To identify key determinants of C-section delivery using statistical methods (Chi-square, GEE).
- To develop predictive models for C-section prediction.
- To interpret model outcomes using SHAP explainability techniques.
- To assess inequality and variation across socioeconomic groups in C-section utilization



Data Source: Bangladesh Demographic & Health Survey 2022

Conducted by NIPOORT & ICF International | Stratified two-stage cluster sampling design | Data authorized by DHS Program (USAID), Eligible women aged 15–49 with a live birth in the 3 years before the survey



Survey Design: Stratified two-stage cluster sampling

Stage 1: PSUs (Enumeration Areas) selected by PPS sampling

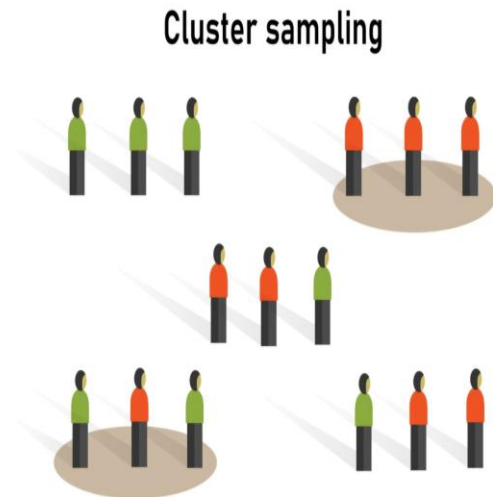
Stage 2: Households selected via systematic random sampling

Why use survey weights?

BDHS uses a complex cluster-sampling design. Without weights, estimates would be biased toward over-sampled groups. Weights ensure results represent all 170 million Bangladeshis.

Why 5,052 not 64,722?

59,670 records excluded: missing outcome variable, survey design variables (PSU, stratum, weight), or key covariates. This ensures complete, unbiased analysis.



Variables Included in the Analysis

OUTCOME VARIABLE

Mode of Delivery
(C-Section=1, Normal=0)

Maternal

- ▶ Maternal age
- ▶ Maternal education
- ▶ Employment status

Household

- ▶ Wealth index
- ▶ Place of residence
- ▶ Geographic region

Obstetric

- ▶ Birth order (parity)
- ▶ ANC visits
- ▶ Child size
- ▶ Birth type

Healthcare

- ▶ Delivery Place
- ▶ Husband's education
- ▶ Husband's occupation

Survey Design

- ▶ Primary Sampling Unit
- ▶ Sample Weight
- ▶ Stratum

* Delivery facility type (Home/Public/Private/NGO) was the STRONGEST predictor in bivariate analysis (Cramér's $V = 0.763$). However, it was EXCLUDED from GEE due to quasi-complete separation no C-sections occurred in home deliveries. It was retained in ML analysis where this is not an issue.

Integrated Analytical Framework

Statistical inference + Machine learning + Explainable AI

Stage 1 Statistical Inference

- 1 Chi-square bivariate tests
- 2 Cramér's V effect size
- 3 Survey-Weighted GEE model
- 4 Adjusted Odds Ratios (AOR)

Stage 2 Machine Learning

- 1 Ensemble feature selection
- 2 Hyperparameter tuning
- 3 6 classifiers trained & tested
- 4 Youden's J threshold optimization
- 5 Ensemble Stacking model

Stage 3 Explainable AI (SHAP)

- 1 Global feature importance
- 2 Individual-level waterfall plots
- 3 Dependence & interaction plots
- 4 Socioeconomic inequality analysis

 **Key Principle: All three stages complement each other where GEE provides inference, ML provides prediction and SHAP provides explanation.**

Survey-Weighted GEE for Population Representative Inference



The Challenge with Survey Data

BDHS uses cluster sampling women from the same village are more similar to each other than to women from different villages **which Violation of independence assumption.**

Use Generalized Estimating Equations (GEE) Because:

1. Clustered Data Adjustment

GEE accounts for intra-cluster correlation in BDHS data, unlike standard logistic regression which assumes independence.

2. Population-Averaged Effects

GEE estimates marginal effects, making it suitable for assessing overall socioeconomic and demographic influences at the population level.

3. Robust Inference

It provides sandwich (Huber–White) robust standard errors, preventing biased p-values and false significance due to clustering.

And Standard logistic regression would give biased estimates So, findings reflect the real national picture.

The Findings of Descriptive Analysis

Weighted descriptive statistics 5,052 women, representing Bangladesh

44.65%

**C-section prevalence
(weighted)**

Max % of

45%

**Delivered in
private facilities**

Max % of

52.5%

1-3 ANC Visit

Max % of

54.4%

**Mother Education
Secondary**

Max % of

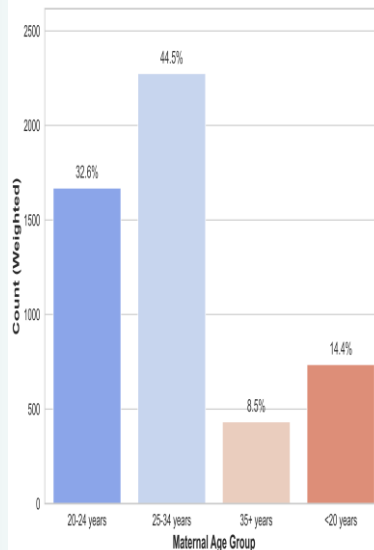
24.5%

Region (Dhaka)

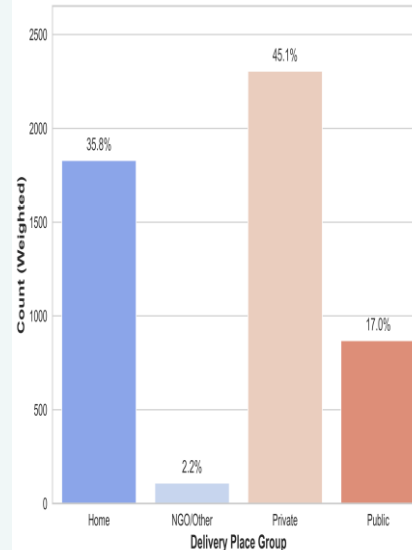
Maternal Profile

- **Most common age group:**
25–34 years (44.49%)
- **Secondary education:**
54.40% of mothers
- **Not employed:**
77.64% of mothers
- **First births:**
37.86% of deliveries
- **Multiple births:** only 0.95%

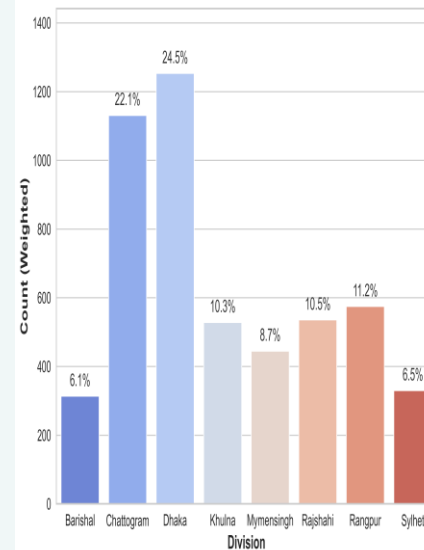
Maternal Age Group Distribution



Delivery Place Group Distribution



Division Distribution



Bivariate Associations with Cesarean Section Delivery

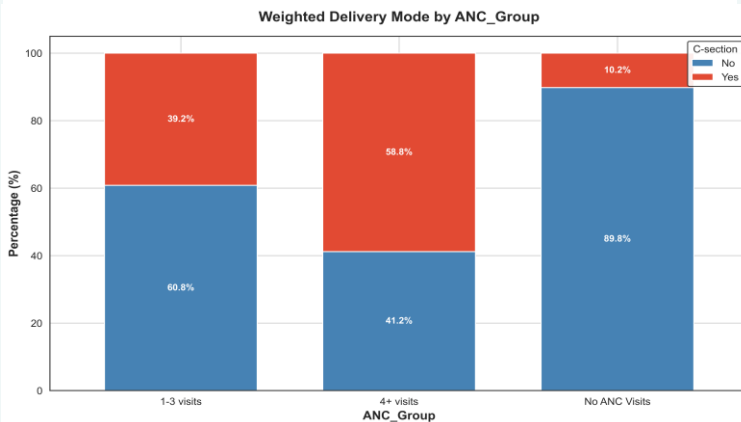
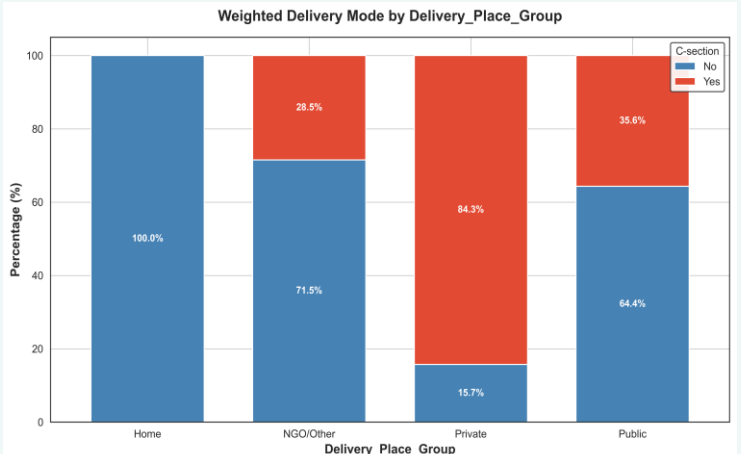
Chi-square tests + Cramér's V effect sizes all significant predictors

Strongest Predictor: Privet Delivery Place

$\chi^2 = 2,979.88$ | $p < 0.001$ | Cramér's V = 0.763 (Very Strong)

Private: 84.3% vs Public: 35.6% vs Home: 0%

Variable	Cramér's V	Sig.	Association	Key Finding
Wealth Index	0.299	***	Strong	Richest: 67.8% vs Poorest: 22.8%
Maternal Education	0.263	***	Strong	Higher edu: 67.6% vs None: 24.2%
ANC Visits	.277	***	Strong	4+ visits: 58.8% vs No ANC: 10.2%
Birth Order	0.206	***	Moderate	Firstborn: 51.2% vs 4th+: 21.1%
Region	0.241	***	Moderate	Khulna: 66.2% vs Sylhet: 27.8%
Maternal Age	0.034	NS	Negligible	Not significant (p = 0.111)



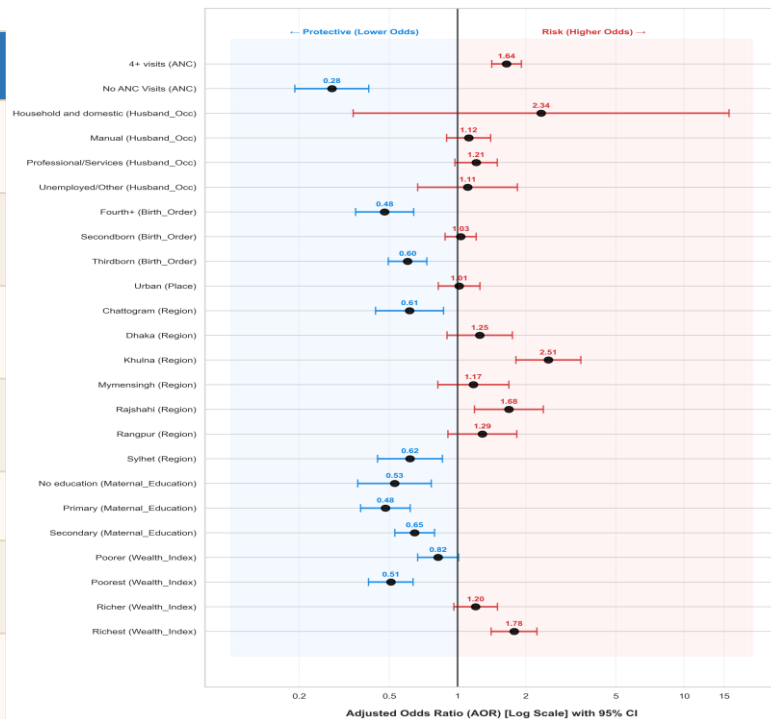
Adjusted Effects of Key Determinants

Survey-Weighted GEE estimates the adjusted effect of each predictor while accounting for other variables.

After adjusting for other variables, several factors remained significant.

Factor	Finding	What it means
4+ antenatal visits	AOR = 1.64 ↑	64% higher chance of C-section vs. fewer visits
No antenatal visits	AOR = 0.28 ↓	72% lower chance, mainly because they delivered at home
Richest wealth group	AOR = 1.78 ↑	78% higher chance vs. middle-wealth women
Poorest wealth group	AOR = 0.51 ↓	Half the chance of C-section vs. middle-wealth women
4th birth or more	AOR = 0.48 ↓	52% lower chance vs. first birth
Higher maternal education	AOR = highest	Most educated women have the highest C-section odds
Khulna Division	AOR = 2.51 ↑	2.5 times higher odds than Barishal.

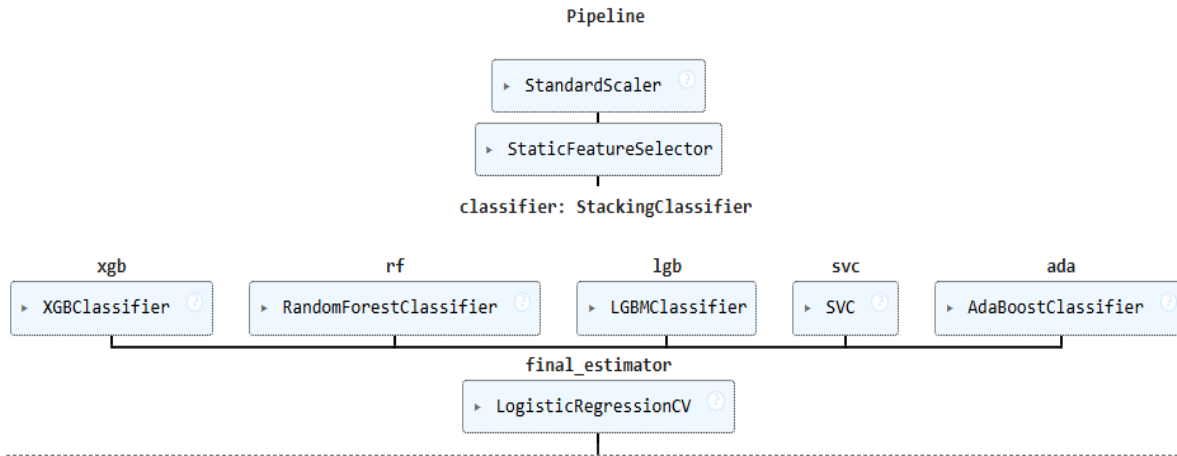
Forest Plot: Socio-Economic Determinants of C-Section Delivery (GEE Model Automatic Results)



Key Point: Wealth, antenatal care, birth order, and education all independently affect C-section odds, even after controlling for other factors.

Machine Learning : How Built and Validated Models

After identifying the significant factors, we developed six machine learning models to predict C section.



STACKING ENSEMBLE (Primary ML Model)

Meta-classifier: Logistic Regression with L2 regularization, trained on out-of-fold predictions from all 5 base learners. Leverages complementary strengths of tree-based, kernel-based, and linear models.

Modeling pipeline

1

Ensemble Feature selection

2

Cross Validation and Hyperparameter tuning

3

Youden's J threshold

4

Survey-weighted evaluation

Pipeline: Leakage-free | 80/20 stratified split | 5-fold stratified cross-validation | Youden's J threshold | Survey weights applied throughout

Performance Comparison Across ML Models

Best Overall Model: Stacking Ensemble

F1-Score: 87.06% | ROC-AUC: 0.924 | Recall: 91.22% | Accuracy: 87.81%

Model	Threshold	Accuracy	Recall	Precision	F1-Score	ROC-AUC
Logistic Regression	0.388	87.59%	91.38%	82.79%	86.88%	0.925
Stacking Ensemble	0.335	87.81%	91.22%	83.26%	87.06%	0.924
Random Forest	0.412	87.95%	89.84%	84.37%	87.02%	0.922
AdaBoost	0.384	87.40%	90.53%	82.99%	86.59%	0.922
XGBoost	0.502	87.78%	88.66%	84.83%	86.70%	0.916
SVM	0.450	87.98%	90.06%	84.28%	87.07%	0.912

High ROC-AUC values (0.91–0.93) suggest that cesarean section delivery is highly predictable from socioeconomic and healthcare characteristics.

Model performance

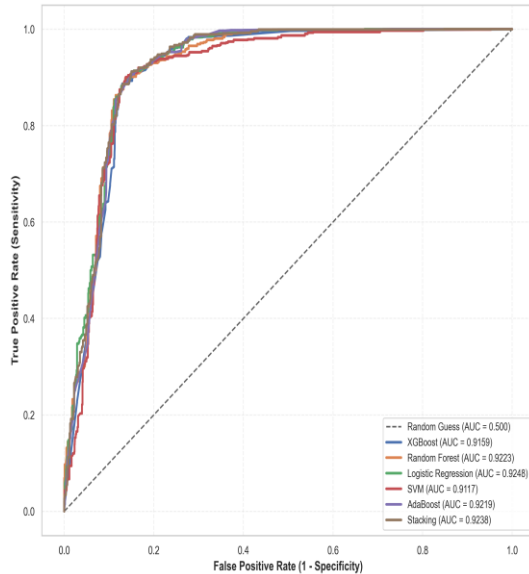
Overall, all models performed well, but the Stacking Ensemble provided the best balance between accuracy, recall, and overall prediction performance

Stacking
ROC-AUC **0.924**

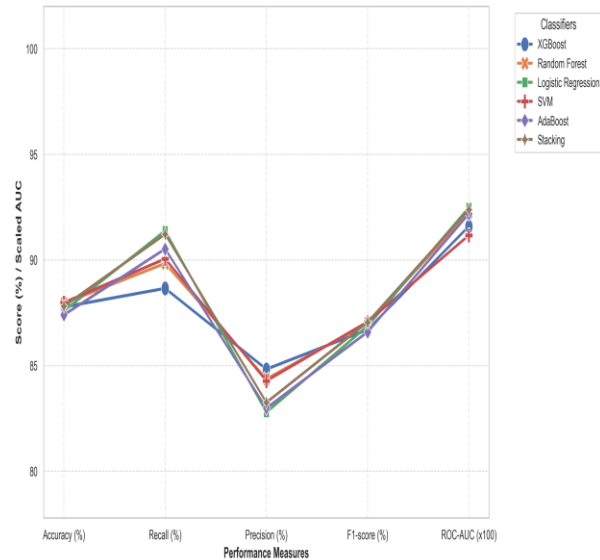
Stacking
F1-score **87.06%**

Stacking
Recall **91.22%**

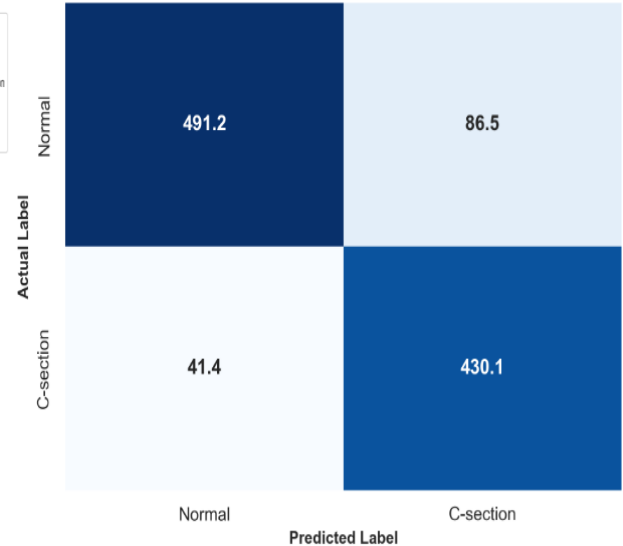
ROC Curve Comparison of Survey-Weighted Models



Connectivity Plot: Comprehensive Model Performance Comparison



Confusion Matrix: Stacking
(Threshold: 0.335)



Model Reliability: Calibration, Stability & Robustness Tests

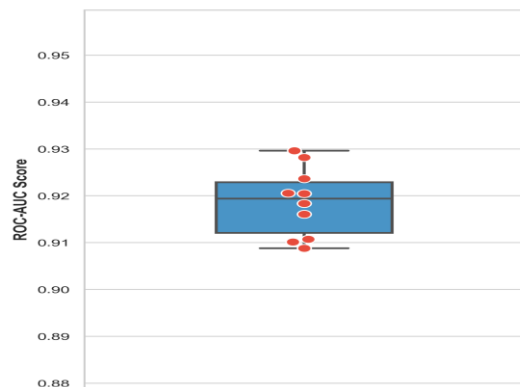
We didn't just build models, we stress-tested them.

Cross-Validation Stability



Mean ROC-AUC = 0.9187
SD = 0.0069

10-Fold Cross-Validation Stability
(Final Stacking Model)



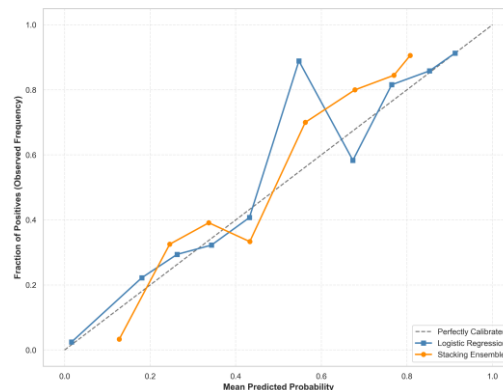
10-fold cross-validation shows consistently high performance across all data splits. Low standard deviation confirms stable, reliable results.

Calibration Analysis



Stacking Ensemble
better calibrated

Reliability Diagram: Logistic vs. Stacking Model Calibration



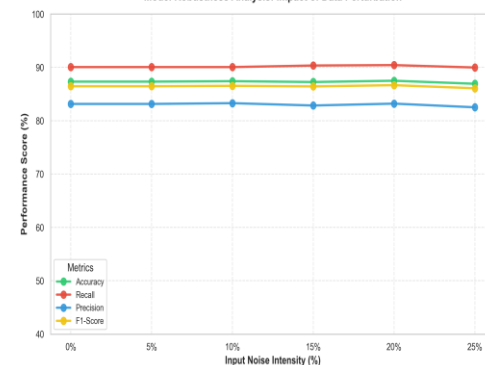
Calibration curves confirm predicted probabilities align closely with observed outcomes. Reliable for risk-based clinical decision support.

Gaussian Noise Robustness



Stable up to 20%
noise injection

Model Robustness Analysis: Impact of Data Perturbation



We introduced 0–25% random noise to test data. Performance barely changed up to 20% noise. Model is robust to real-world data imperfections.

SHAP Analysis: Opening the Black Box with SHAP

SHAP (SHapley Additive exPlanations)

A game-theory based method to fairly distribute each feature's contribution to the model's prediction.

Understanding the SHAP Summary Plot

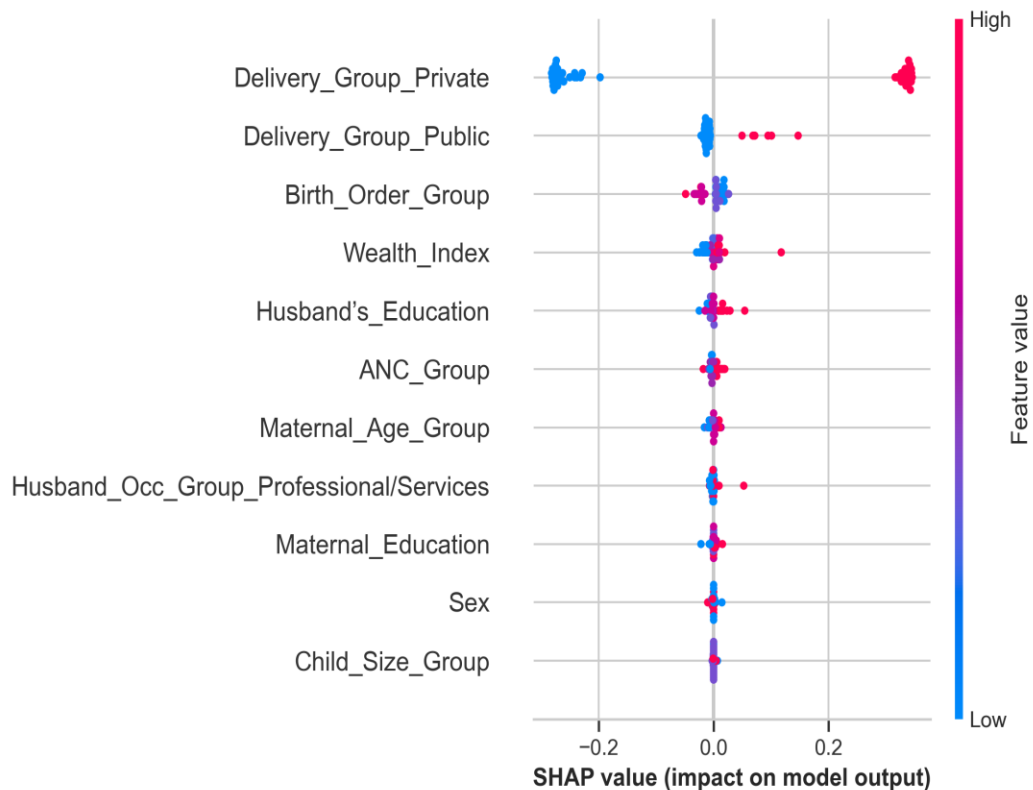
- Red point = Higher feature values
- Blue point = Lower feature values
- Right side = Increases C-section probability
- Left side = Decreases C-section probability



Key Finding

Private delivery place, ANC visit and wealth status increased C-section risk, whereas higher birth order was associated with a lower probability of C-section delivery.

SHAP Summary Plot for Stacking Model Analysis



SHAP Insights: Interaction Between Wealth and Private Facility Delivery

Dependence Plot Shows

Wealth increased C-section risk more strongly among women delivering in private facilities.

Highest Risk Group

Wealthiest women delivering in private facilities showed the highest predicted probability of C-section.

Poorer Women in Private Facilities

Even in private facilities, poorer women have lower C-section rates but still higher than public facilities.

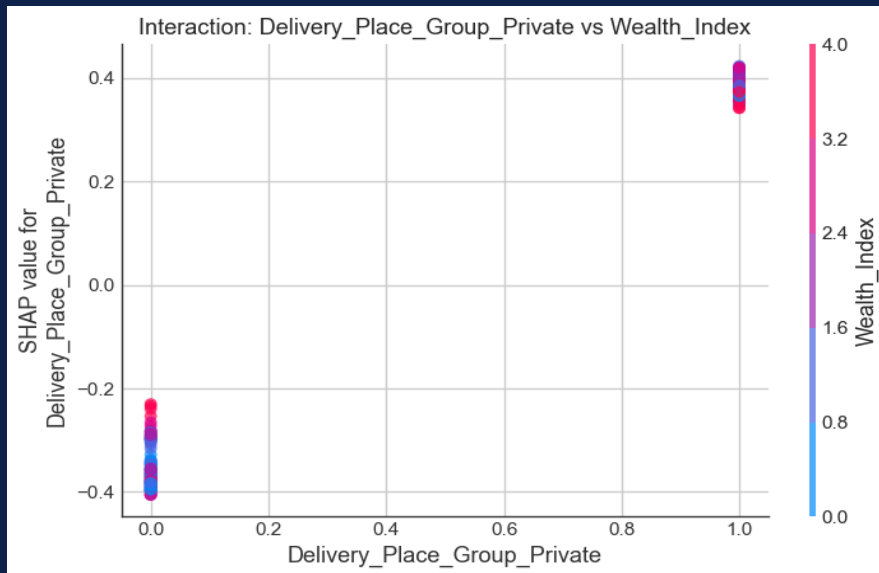
Wealth Effect in Public Facilities

Wealth had little impact on C-section risk in public facilities.

Home Delivery:

Home delivery was consistently associated with normal delivery, with virtually no C-sections observed across all wealth groups.

How SHAP Works for Interaction?



Wealth and private facility together lead to the highest C-section probability.

Waterfall Plot: Individual Prediction (Case 36)

For Case 36, the predicted C-section probability was 79%. Private facility delivery was the primary driver (+34%), whereas education and occupation contributed little to the final prediction.

Base probability $E(f(x))$:

The average baseline probability for a C-section is **43.2%**.

Final prediction $f(x)$:

For this specific case, the predicted probability increases significantly to **0.787** (approx. 79%).

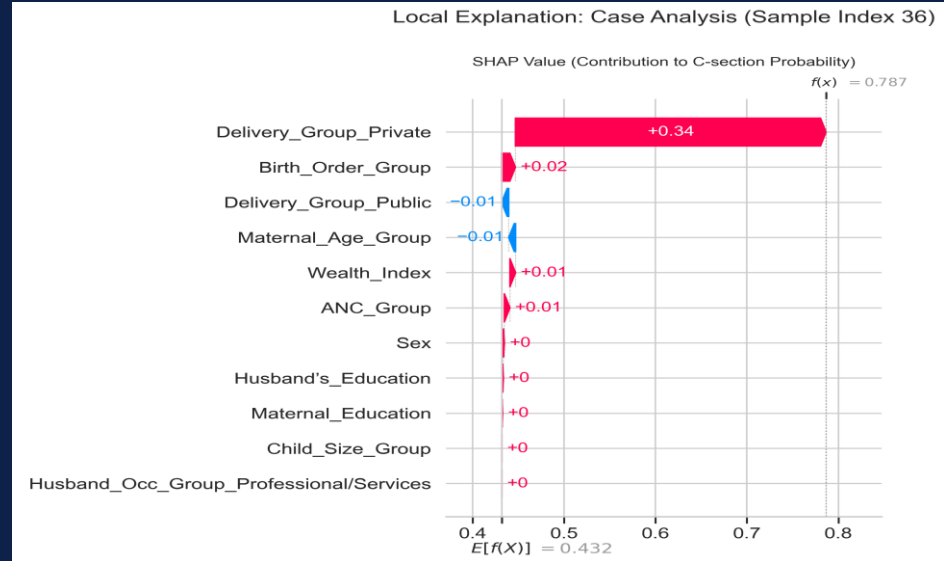
Delivery in Private (+0.34):

The most dominant factor; delivering in a private facility increases the C-section probability by **34%**.

Delivery in Public (-0.01):

Public sector delivery slightly reduces the probability.

Individual Prediction: How SHAP Works for One Woman



Actual outcome in Dataset: C-Section

Model predicted probability = 79% → Predicted: C-Section ✓

This example shows how each factor contributes to one prediction.
Where Private facility plays the biggest role.

Statistical vs Machine Learning: Both Tell the Same Story

Convergence of evidence strengthens our confidence in the findings

An important finding is that both statistical analysis and machine learning identified almost the same key predictors. This consistency strengthens the reliability and confidence in our findings.

Predictor	GEE (Inference)	ML Rank (SHAP)	SHAP Direction	Agreement
Delivery Place	Excluded (complete separation)*	#1 Feature (SHAP = 1.74)	Private facility dominates	✓
Wealth index	Richest: AOR = 1.78***	#4 Feature	Strong positive SHAP	✓
ANC visits	4+ visits: AOR = 1.64***	#6 Feature	Positive contribution	✓
Birth order	4th+: AOR = 0.48***	#3 Feature	Strong negative SHAP	✓
Maternal education	Lower edu: AOR < 1***	#9 Feature	Moderate positive SHAP	✓
Maternal age	NOT significant	Low importance	Near-zero SHAP	✓

Delivery Place was excluded from GEE for statistical reasons, but analyzed in bivariate and ML stages, confirming its overwhelming importance.

Policy Implications: What Bangladesh Must Do

Based on our findings We recommend:



1. Private Facility Regulation

- Strengthen monitoring of private maternity facilities.
- Publicly report hospital C-section rates.
- Ensure evidence-based surgical delivery practices



2. Reorient Antenatal Care

- Integrate delivery-mode counseling into ANC services.
- Promote shared decision-making between women and providers.
- Address misconceptions about C-section safety.



3. Reduce Socioeconomic Inequalities

- Reduce both overuse and underuse of C-sections.
- Improve access to quality obstetric care for disadvantaged groups.
- Strengthen public maternity services.



4. Geographically Targeted Action

- Implement targeted interventions in high-rate regions.
- Monitor low-rate regions to prevent unmet obstetric needs.

Limitations & Future Directions

Despite our robust methodology, there are a few limitations to consider:

⚠ Cross-sectional design

The BDHS dataset is cross-sectional, limiting the ability to establish causal relationships between predictors and C-section delivery.

⚠ Lack of clinical information:

The BDHS dataset does not include detailed clinical indications, limiting the ability to distinguish between medically necessary and non-medically indicated C-sections.

⚠ Exclusion of home delivery in GEE model

Due to complete separation (0% C-section), home deliveries were excluded, which may reduce the comprehensiveness of the regression analysis.

Future Research Directions:

Future research can build on this work by incorporating longitudinal data, integrating clinical records, and applying causal machine learning approaches to better understand C-section determinants.

THE CONCLUSION

Cesarean Section in Bangladesh is A Structural Crisis, Not Just a Clinical One

01 **The Scale is Alarming**

At 44.65%, Bangladesh's C-section rate is 3-4× the WHO threshold. This is not a clinical pattern. It is a public health crisis driven by structural forces.

02 **The Driver is Clear**

Delivery facility type dominates all analytical approaches. Private facility attendance is the single strongest predictor, amplified by household wealth. This points to supplier-induced demand and market failures in maternity care.

03 **The Solution is Structural**

Individual counseling alone will not solve this. Coordinated regulatory reform, ANC reorientation, and equity-focused strategies are needed to bend the curve on unnecessary C-sections.

Without structural reform, the current trajectory will exacerbate health inequities and undermine evidence-based maternal care in Bangladesh.



Questions & Discussion

I would be happy to take any questions.



Quick Reference: Key Numbers

- ▶ Data: BDHS 2022 | n = 5,052 | Method: GEE + ML + SHAP | C-section prevalence = 44.65%
- ▶ Top predictor: **Private facility delivery (Cramér's V = 0.763; SHAP #1)**
- ▶ Key GEE results: Richest AOR = 1.78 | 4+ ANC visits AOR = 1.64 | 4th+ birth AOR = 0.48
- ▶ Best ML model: Stacking Ensemble where, **F1: 87.06% | ROC-AUC: 0.924 | Recall: 91.22%**
- ▶ Conclusion: C-section is socioeconomic, not just medical structural reform needed.

Key References

Selected sources supporting this analysis

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